MSc in Software Design with Cybersecurity

Initial Project Concept

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Project Title:	***AI Act Compliance for Credit Scoring: Automating FRIA, Cybersecurity and Fairness***
Project Background:	**The EU AI Act requires organisations using high-risk AI systems such as credit scoring to conduct a Fundamental Rights Impact Assessment, ensure cybersecurity robustness, provide explainable outputs, manage risks, and detect bias before deployment. Currently no tool exists that brings all of these obligations together in one place, forcing organisations to use separate tools, follow separate processes, and apply different expertise for each. Existing platforms like OneTrust handle GDPR compliance in isolation and do not produce the specific legal documentation the EU AI Act demands. This leaves banks and financial institutions, particularly smaller ones, struggling to comply and exposed to fines of up to 30 million euros.**
Project Challenge:	**To design and build a unified prototype web application that automatically generates a Fundamental Rights Impact Assessment, a cybersecurity threat model, an explainability report, a risk scoring dashboard, and a bias detection report for credit scoring AI systems, all mapped to specific articles of the EU AI Act.**
Proposed Approach:	**A React based web application will be built with a Python FastAPI backend and a Neo4j knowledge graph storing EU AI Act legal requirements, FRIA obligations, and cybersecurity threats as connected nodes. A step-by-step questionnaire will collect information about the credit scoring AI system, which the knowledge graph will traverse to automatically generate all five outputs mapped to Articles 9, 10, 13, 15, and 27 of the EU AI Act. SHAP will be used for the explainability module, IBM AIF360 for bias detection using fairness metrics including**

	demographic parity and equalised odds, and an ISO 31000 based severity scale for the risk scoring dashboard. The prototype will be evaluated on the German Credit dataset measuring legal requirement coverage, threat detection accuracy, fairness metric outputs, and usability through expert review.